

Per-head evictors (SnapKV, Ada-KV, H2O, TOVA)

Prefill with FLASH-ATTENTION OFF
attention weights must be materialised to be read

Top-K per (layer, head)
each of 128-1024 selectors picks its own K tokens

UNION across heads and layers
a cell survives if ANY selector keeps it

seq_rm marks cells free
survivors are never moved

SPARSE: decode scans to the last live cell



retains 67.7-84.9 % of cells at the same K

μ KV (ours)

Prefill with FLASH-ATTENTION ON
in-graph kq_evict side node scores inside FA

One keep-set per layer
scored on aggregated attention mass, not per head

NO union to take
the selected set is already sequence-level

COMPACTION: state round-trip
survivors moved together; 32 ms - 6.1 s once

DENSE: decode scans K cells, not N



retains 8.7-10.0 % = its budget; keeps q8_0 KV